Water groups and arrays

H2O: Why Water Is Special

Name _____ Date _____

Write one answer in each blank. Use repeated addition to show every equal group or row.

1. Big fills 2 ice trays with 4 ice cubes in each tray. How many ice cubes are there altogether?

$4 + 4 =$ _____

Show your thinking.

2. Little makes 3 rows of 3 water drops on wax paper. How many drops are in the array?

$3 + 3 + 3 =$ _____

Show your thinking.

3. There are 5 cups with 2 paper straws beside each cup. How many paper straws are there altogether?

$2 + 2 + 2 + 2 + 2 =$ _____

Show your thinking.

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4. Big arranges 3 rows of 5 paperclips for a water test. How many paperclips are in the array?

$5 + 5 + 5 =$ _____

Show your thinking.

5. Little sets 4 trays with 4 small bowls of water on each tray. How many bowls are there altogether?

$4 + 4 + 4 + 4 =$ _____

Show your thinking.

6. Big arranges 5 rows of 4 cups of water. How many cups are in the array?

$4 + 4 + 4 + 4 + 4 =$ _____

Show your thinking.

Three atoms

Every bit of water is made of molecules. Each H2O molecule has two hydrogen atoms joined to one oxygen atom.

Three states

The same water can be solid ice, flowing liquid, or invisible water vapor. Melting and freezing change its state.

Drops gather

Water molecules attract one another. At the surface, that pull gathers drops into rounded beads that can join together.

Surface tension

Surface tension makes the top of water resist being pulled apart. A gently placed paperclip can rest there without sinking.

Roomier ice

Ice molecules form a roomier pattern. Ice is lighter for its size than liquid water, so it floats on top.

Water dissolves

Water can surround and carry away tiny pieces of salt or sugar. That is called dissolving, but not everything dissolves.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

WONDER DAILY

H2O: Why Water Is Special

Write what water can do

H2O: Why Water Is Special · Curiosity day

Name _____ Date _____

Choose a clue. Say your idea first, then draw or write it.



Three atoms together



Drops pull together



Ice floats high

Draw one water clue and add a small arrow to the part you want a reader to notice.

water · molecule · atom · hydrogen · oxygen · ice · melt · freeze · vapor · surface tension · float · dissolve

One water molecule has … dictate the number and kinds of atoms

A drop beads up because … dictate what the molecules do

Ice floats because … dictate how ice compares with liquid water

Does your sentence name a water clue and explain why it happens?

H2O: Why Water Is Special

01 What does H2O mean?



Two hydrogen atoms and one oxygen atom join to make one water molecule.

Look closer. Find the small three-part model beside the glass: two matching pieces and one larger piece.

A little explanation. Water is made of tiny pieces called molecules. Each water molecule is H2O: two hydrogen atoms joined to one oxygen atom. Atoms are even smaller pieces that can join to make molecules.

Try this. Put one ice cube on a plate and start a timer. Use only cool water, keep a towel nearby, and wipe up every spill. Watch until the ice melts.

02 Why do water drops bead up?



The attraction between water molecules helps drops gather into rounded beads.

Look closer. Look at the rounded drops on wax paper and the larger drop where two small ones have joined.

A little explanation. Water molecules attract one another. At the surface, their pull creates surface tension, which helps a drop gather into a rounded bead instead of spreading flat.

Try this. Use a dropper or the end of a straw to place a few water drops on wax paper. Nudge two drops together and watch them join.

03 How can a paperclip rest on water?



Surface tension lets gently placed paperclips rest on the top of water.

Look closer. See how the paperclip rests across the surface while the water bends slightly beneath it.

A little explanation. Surface tension makes the top of the water resist being pulled apart. It can act a little like a skin, but it is not a real solid skin. A paperclip can rest there if it is placed gently.

Try this. Lay a small piece of tissue on water, place a clean paperclip on the tissue, and wait for the tissue to sink while the paperclip stays.

04 Why does ice float?



Ice is lighter for its size than liquid water, so it floats.

Look closer. Compare the ice cube above the waterline with the liquid below it and the winter pond beyond the window.

A little explanation. When water freezes, its molecules settle into a roomier pattern. Ice is lighter for its size than liquid water, so it floats. That unusual clue helps ponds freeze from the top while liquid water remains below.

Try this. Drop an ice cube into a clear cup of water. Look from the side and notice how part of the cube stays above the surface.